



AWS Cloud Day

Prague

ISV203

Modern AI-infused applications for financial markets

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Solutions Architect

Amazon Web Services

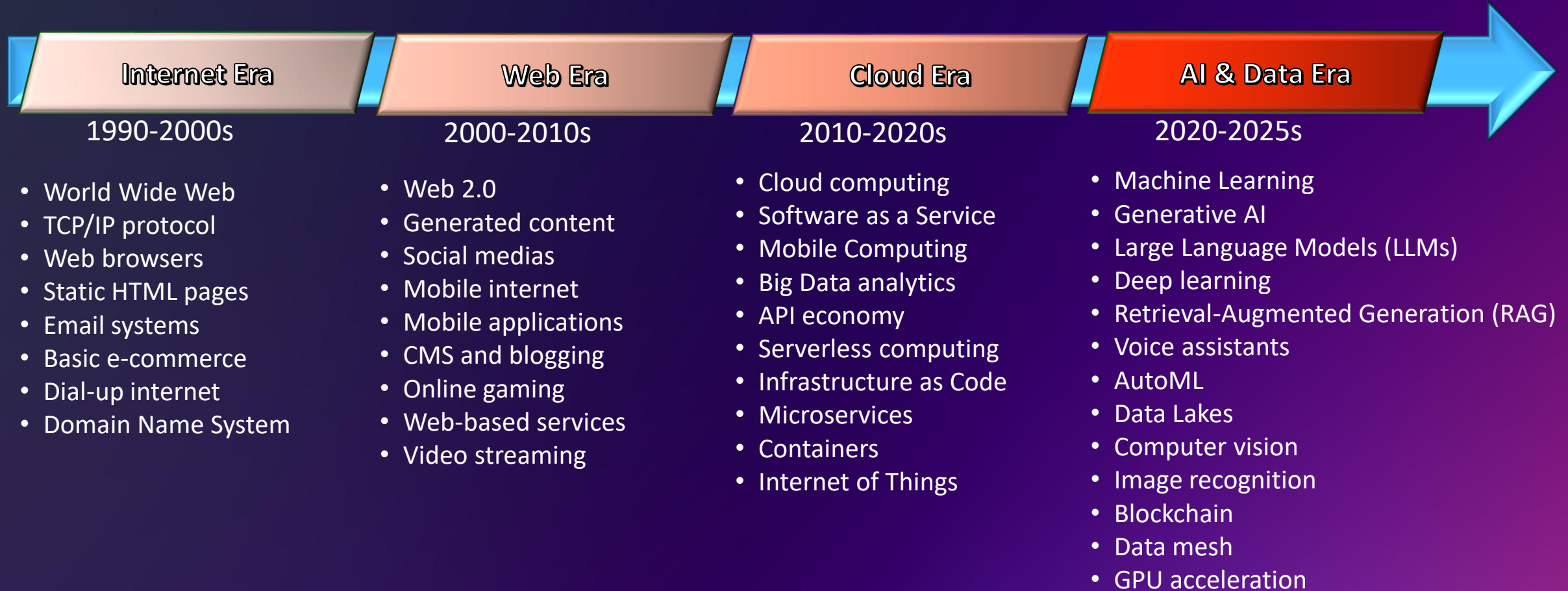
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dxFeed Solutions DE GmbH



IT Evolution



IT Evolution



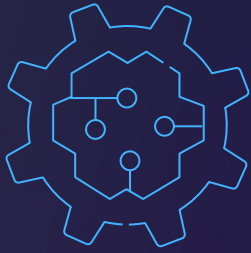
- Autonomous Systems
- Multi-agent systems and coordination
- Multi-modal data integration and analysis
- Advanced reasoning chains and logic
- Autonomous decision-making systems
- LLM orchestration and workflow automation
- Edge Computing
- AI orchestration and management platforms
- Quantum-AI hybrid systems



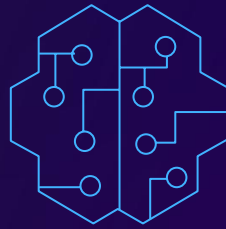
What are AI Agents?

Autonomous software systems that leverage AI to reason, plan, and complete tasks on behalf of humans or systems

Why Agentic AI?



Autonomous
software systems

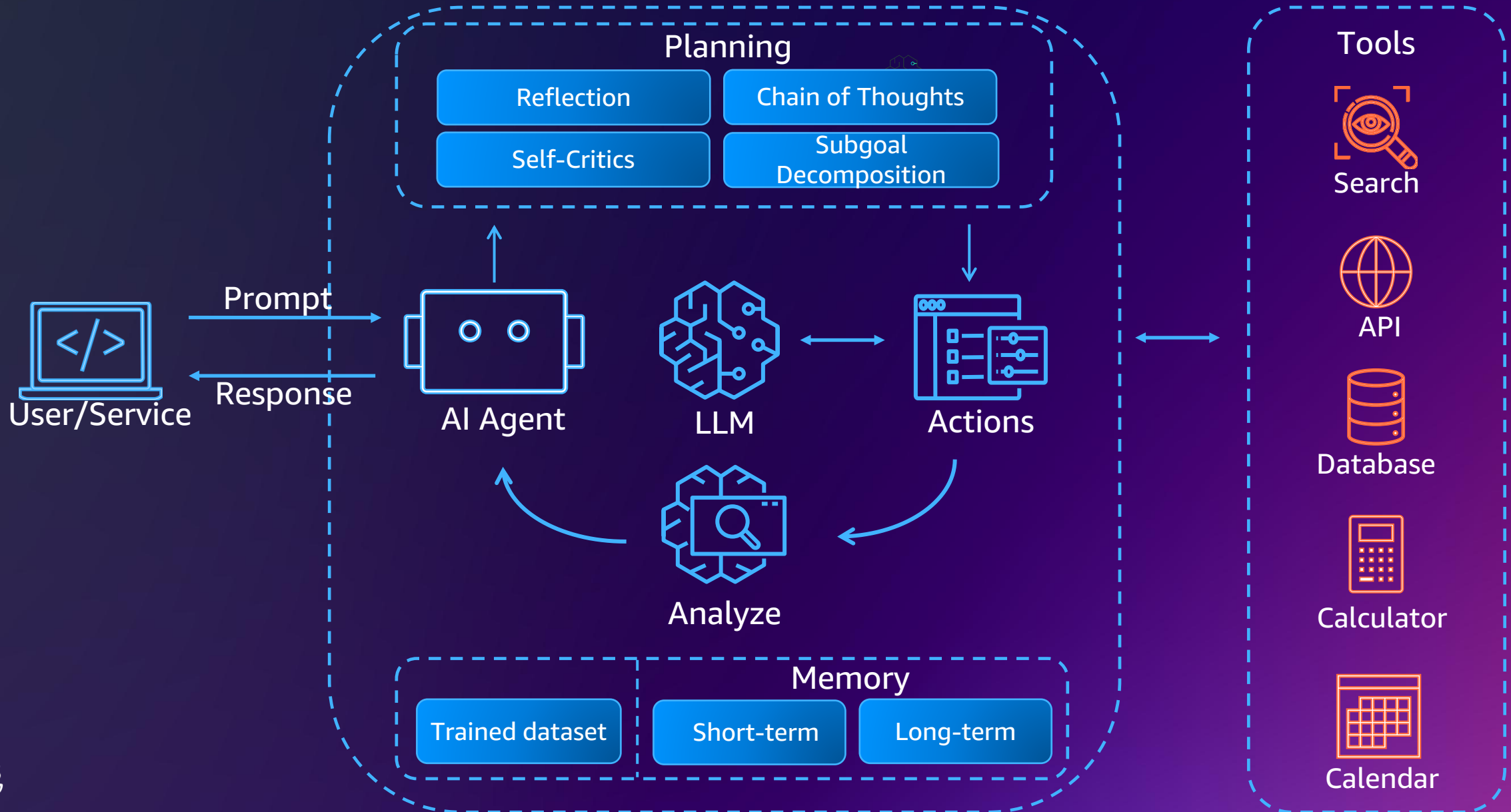


Ability to think
iteratively



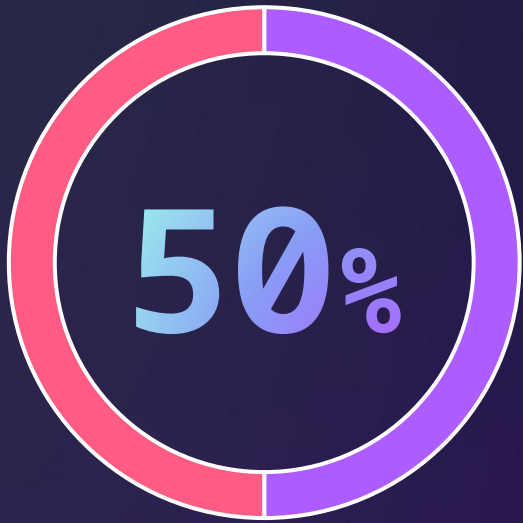
Solve real-world
problems

How AI Agents Work

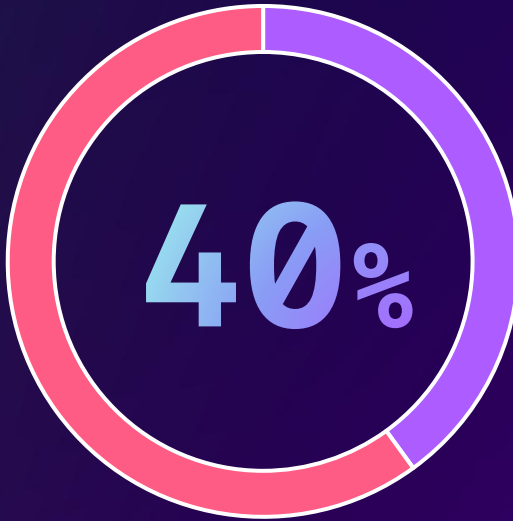


Agentic AI will Transform the Enterprise

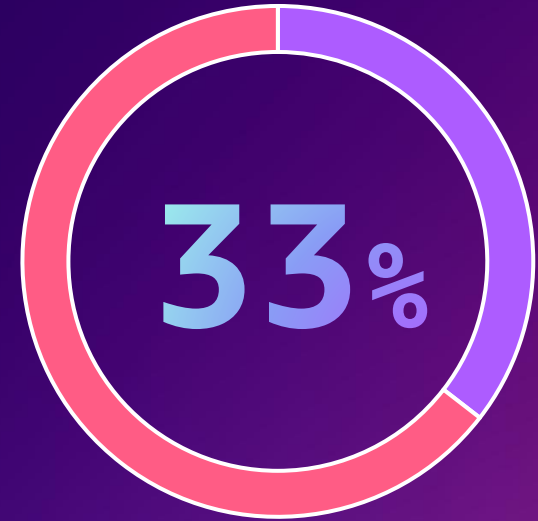
AI Agents will drive performance gains, better problem solving and faster time to action



Organizations will use agents to achieve business value from AI by 2025



Improvements in cycle times, with improved quality, with AI agents



Of enterprise software applications will include agentic AI by 2028

Using Agents in financial markets

dxFeed Unveils Grenadier: Real-Time Anomaly Detection on Order Book Data



dxFeed SummerFox Wins
“Best AI Technology – Trading”



water technology
Sell-Side Technology
Awards Winner 2025
Best artificial intelligence (AI)
technology – Trading
dxFeed

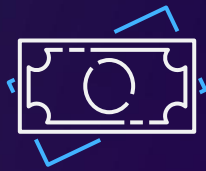


The AI Market Analysis Challenge

Unique demands of financial markets:



Real-time processing
requirements



Low tolerance for
errors



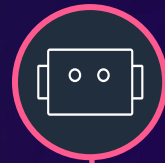
Multiple, often
conflicting data
sources



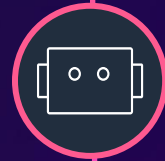
Complex
interdependencies
between data points

The AI Market Analysis Challenge

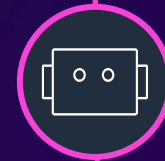
Common pitfalls of AI implementation



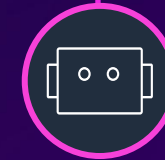
Over-reliance on black-box models



Insufficient data validation



Poor integration with existing systems



Neglecting calculation accuracy

The AI Market Analysis Challenge

What is the square root of
66,777,278,569?



Llama-3.3-70B

258,275



DeepSeek-V3

258,387



258,413



The AI Market Analysis Challenge

What is the highest stock price of AMZN on 22 Aug 2023?



Cannot answer



What is the company under ticker "PERF"?

Perfumania Holdings



The AI Market Analysis Challenge

What is the highest stock price of AMZN on 22 Aug 2023?



\$137.98 per share



What is the company under ticker "PERF"?

Perficient, Inc.



The AI Market Analysis Challenge

What is the highest stock price of AMZN on 22 Aug 2023?



\$135.65 per share



What is the company under ticker "PERF"?

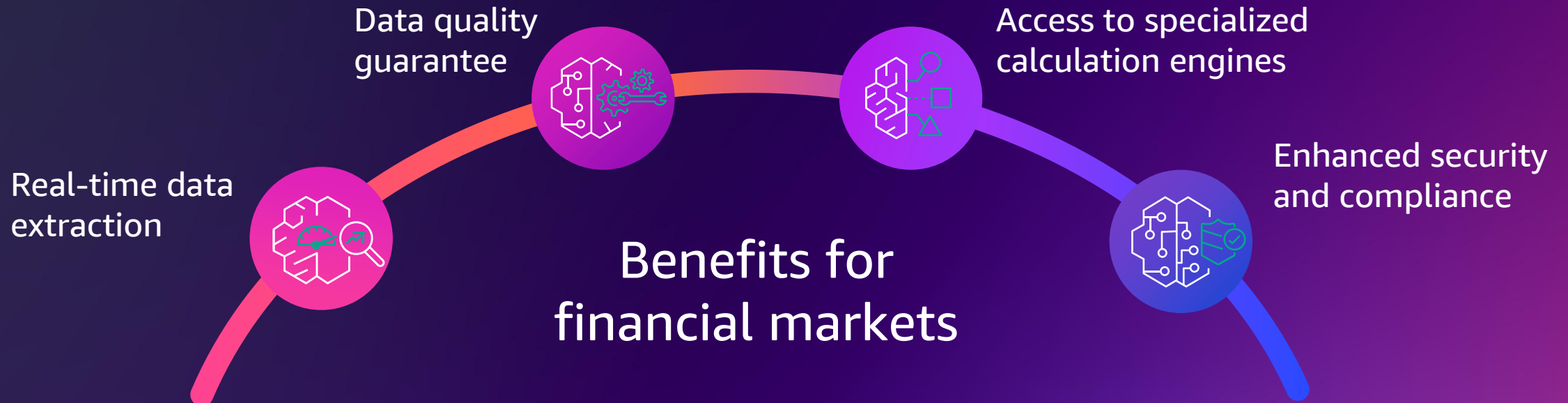
Perfect Corp
(class A shares,
listed on NYSE)



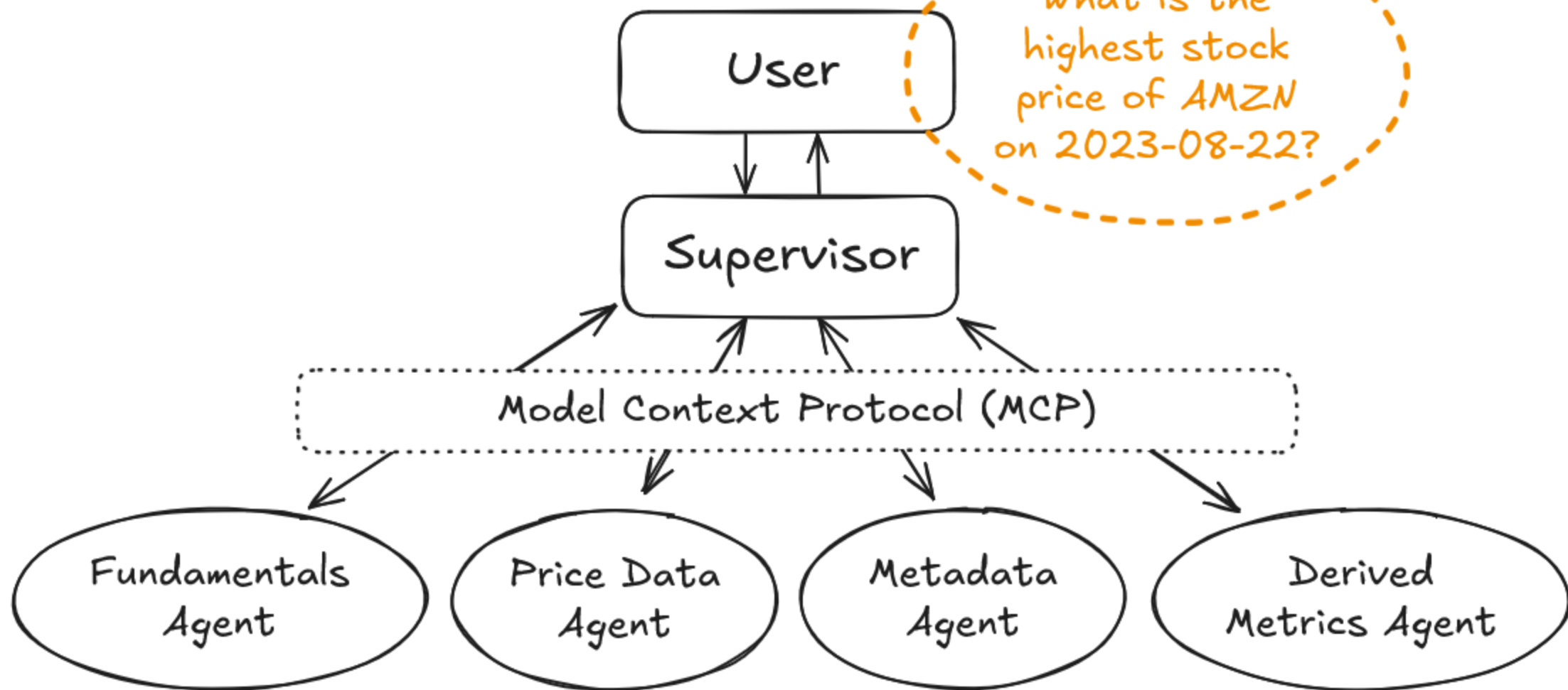
Agentic Architecture via MCP

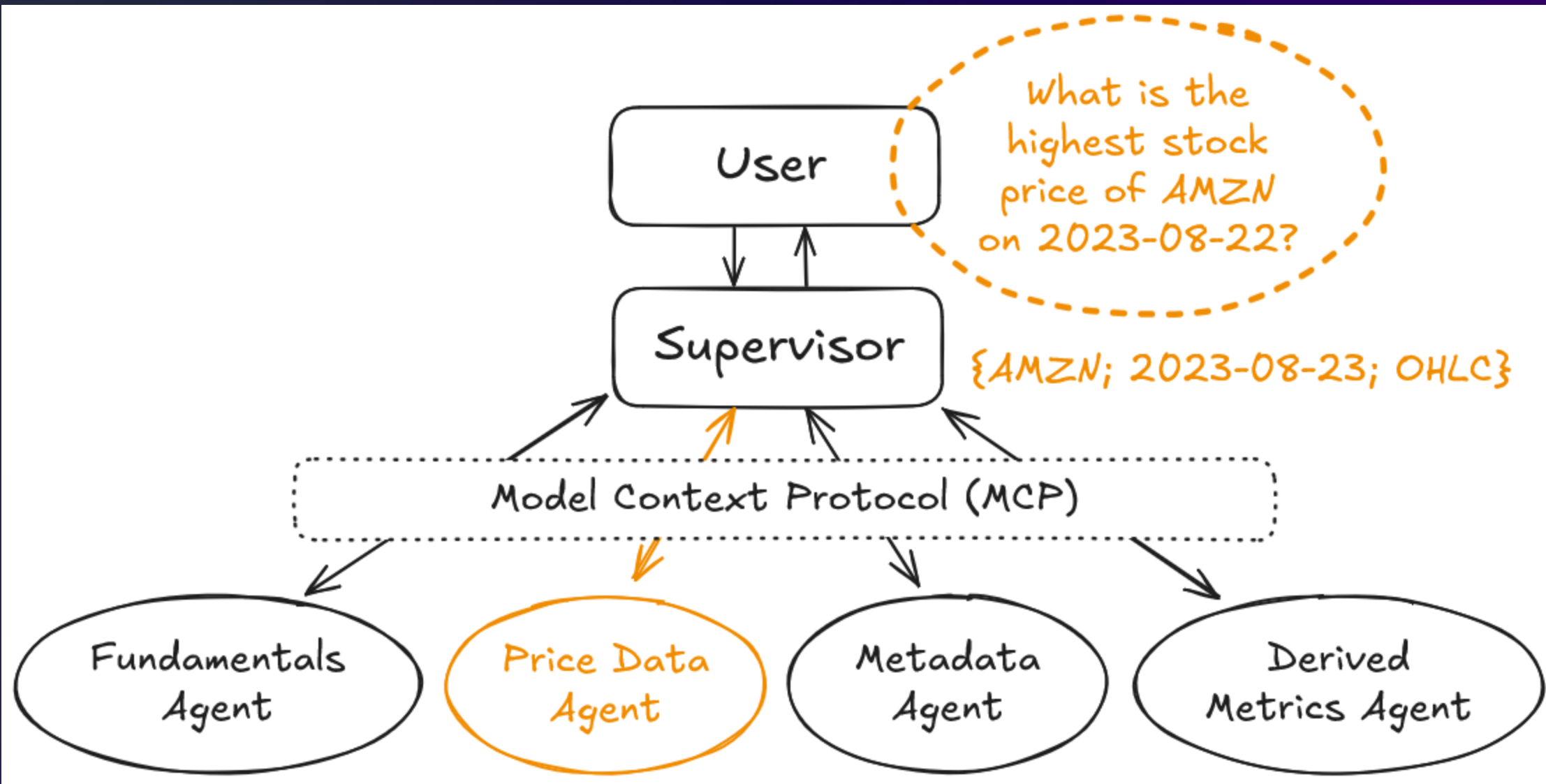
Model Context Protocol (MCP): an open-source standard for connecting AI applications to external systems

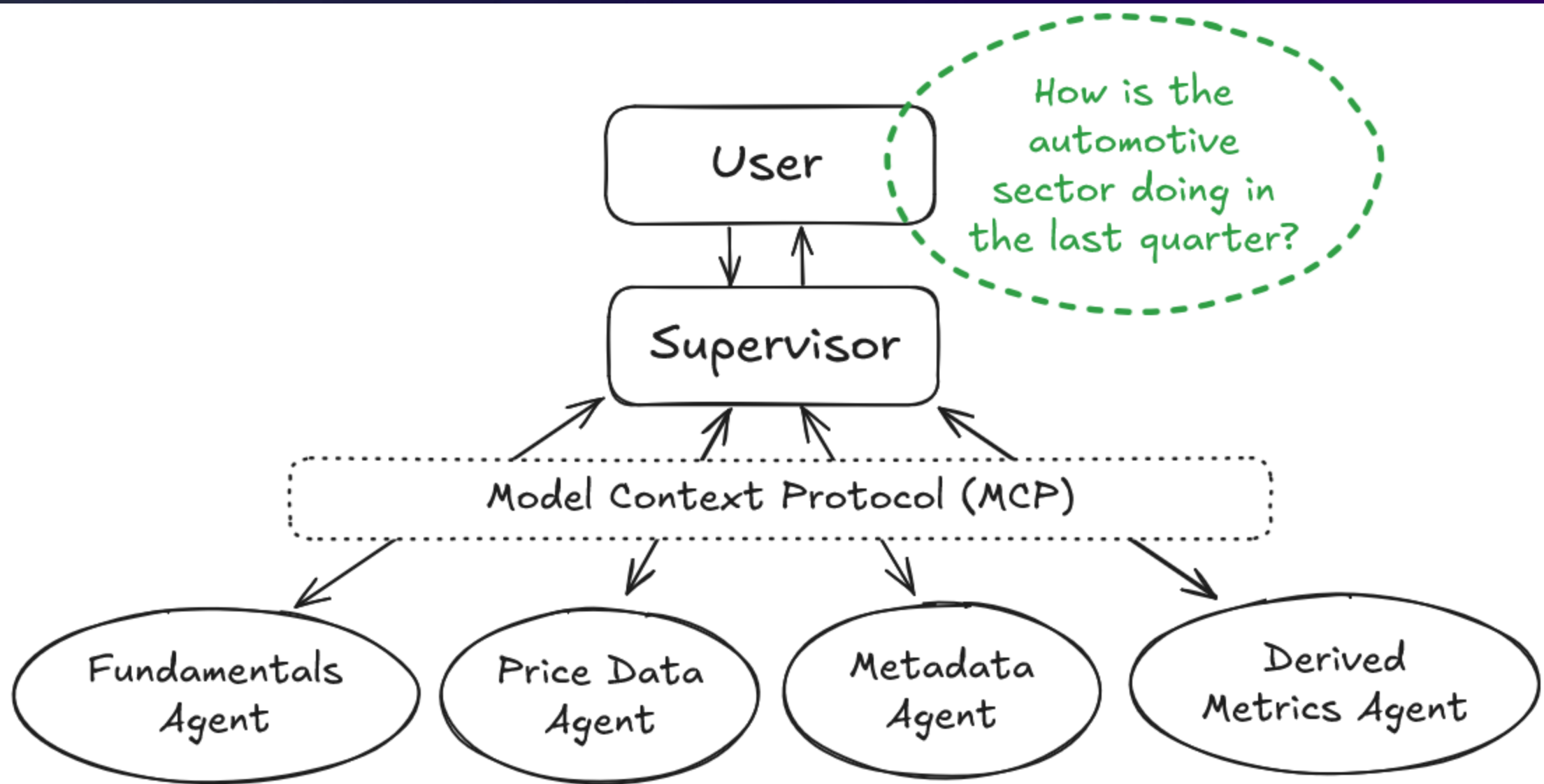
“...like a USB-C port for AI applications”

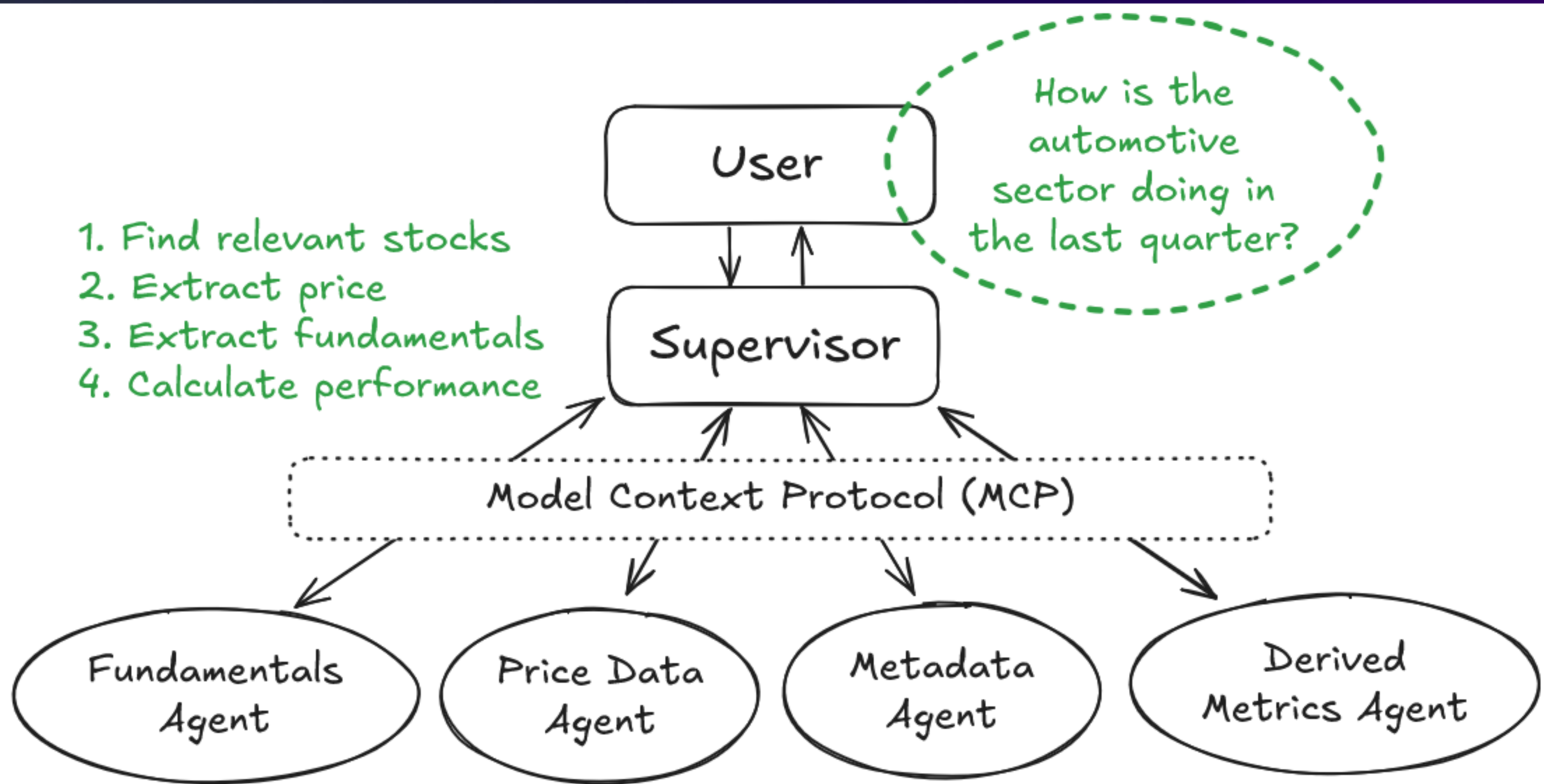


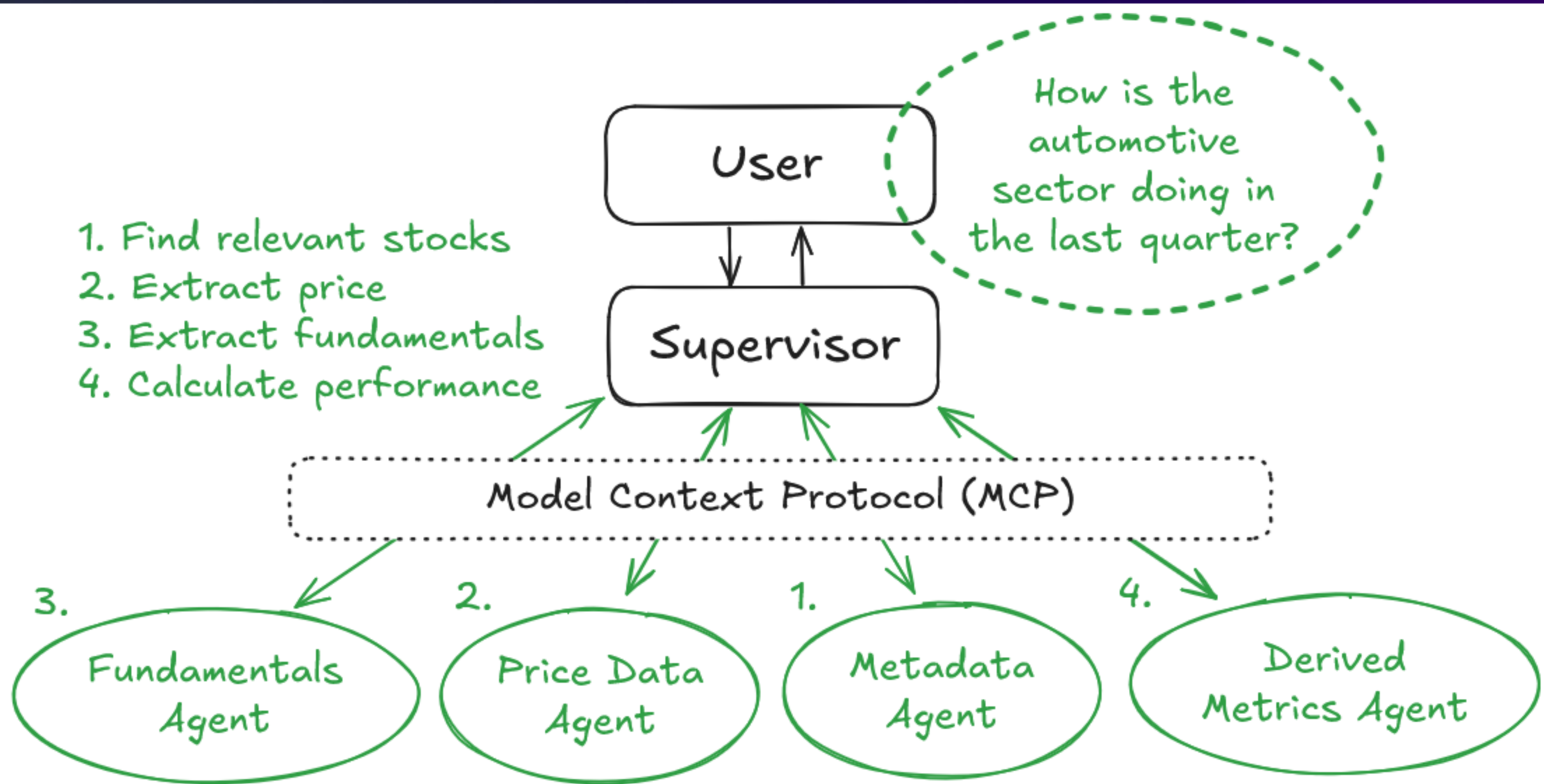
What is the
highest stock
price of AMZN
on 2023-08-22?

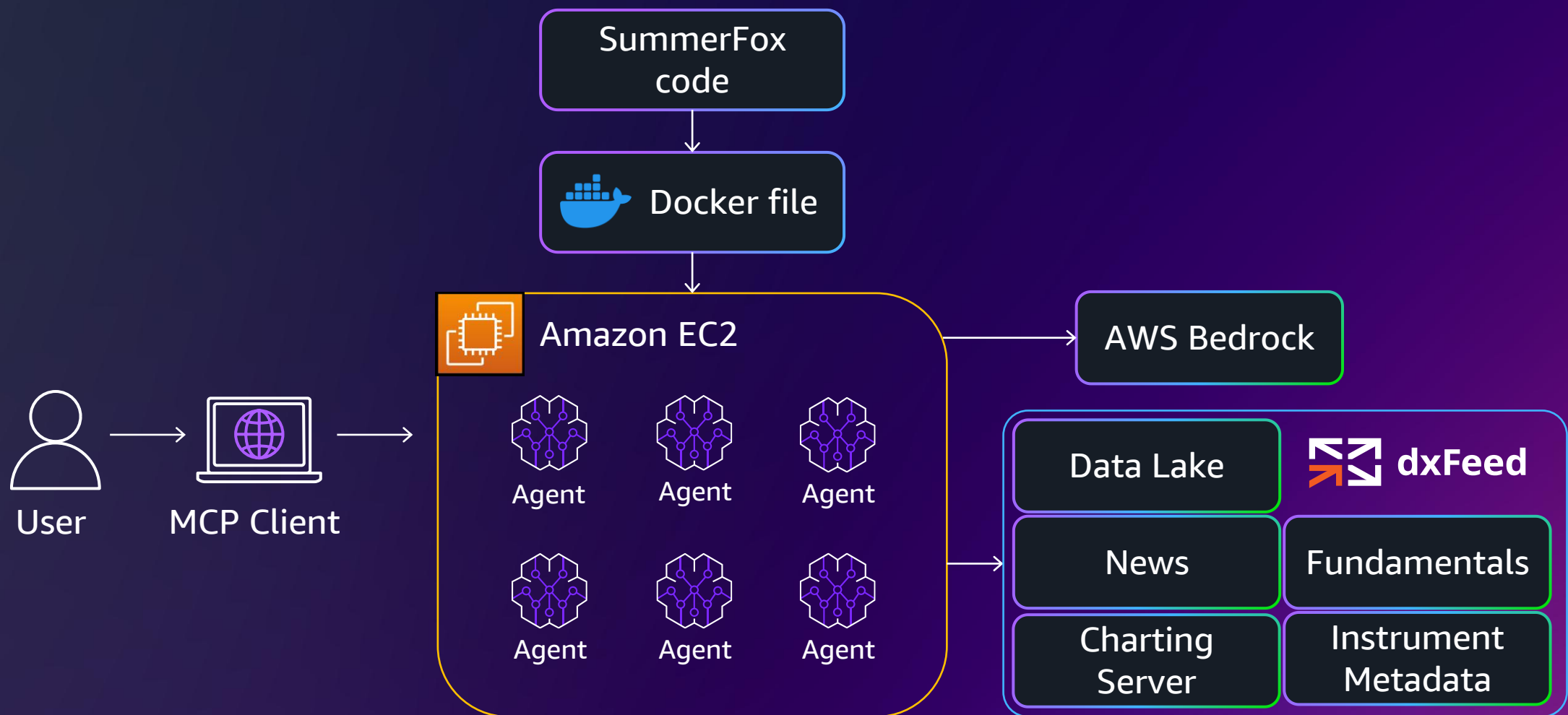


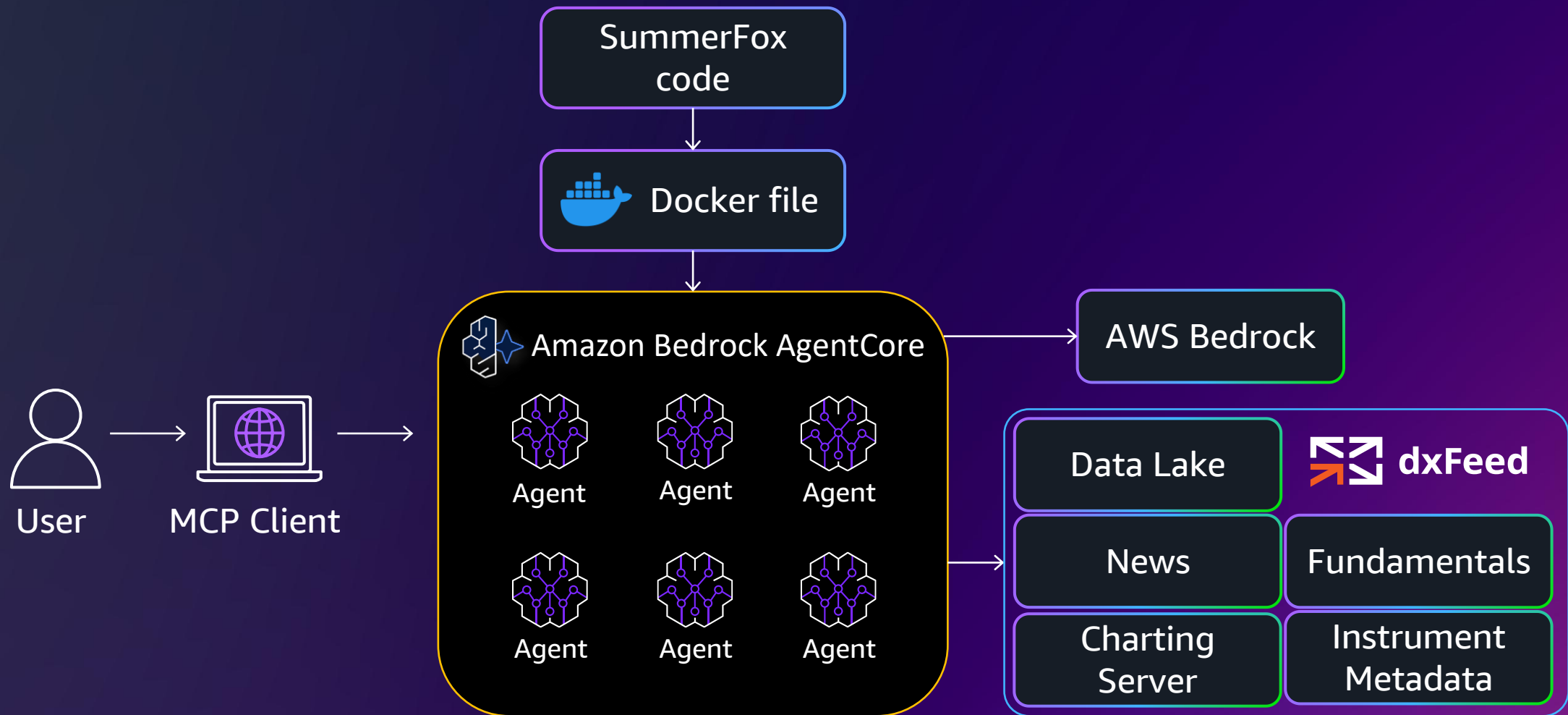








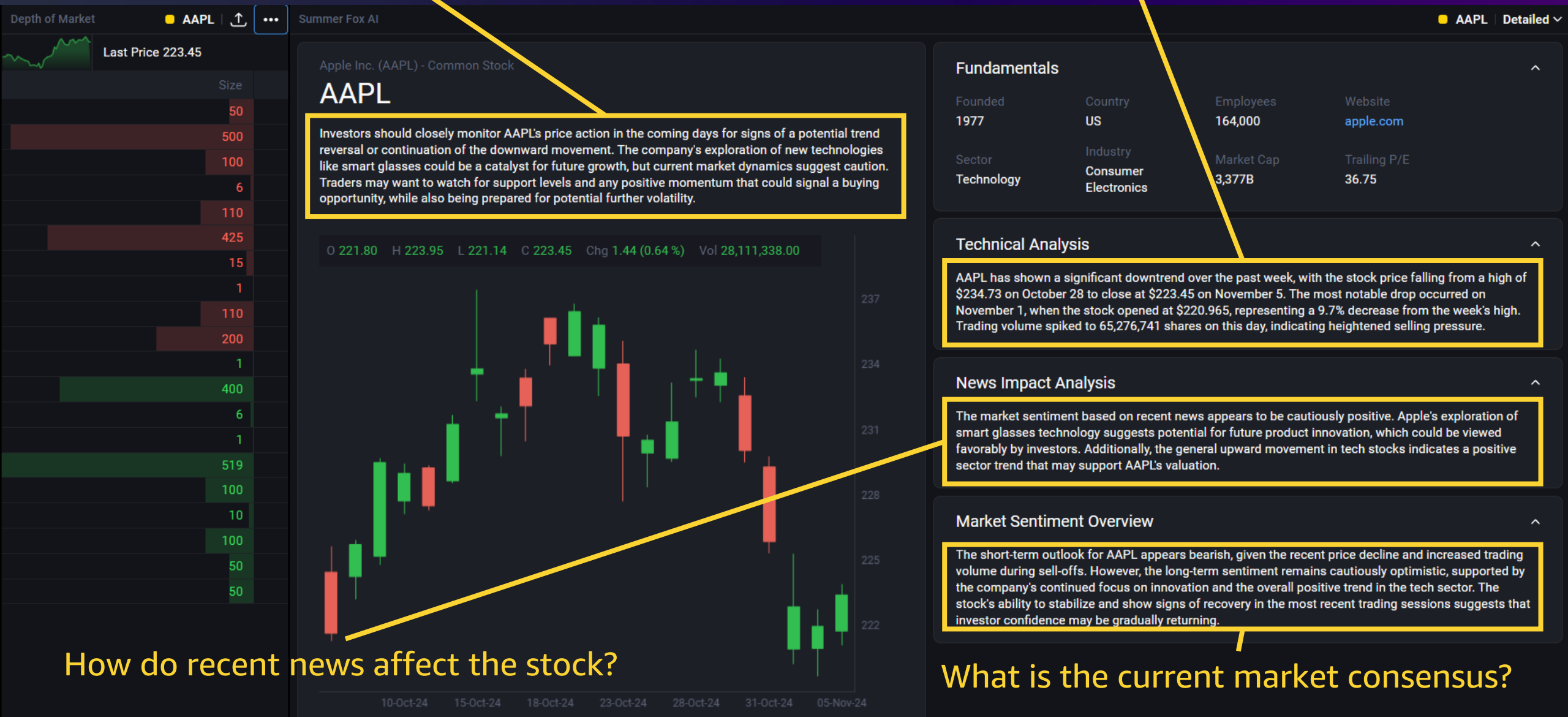




dxFeed SummerFox in Action

What is the high-level outlook?

What data-driven insights are there?



Implementation Considerations

Prompt Engineering Techniques

- Be specific
- Provide few-shot response examples
- Establish specific guardrails
- Know the weak points of an LLM
- Reinforce capabilities with access to tools and agents

Guidelines for Production

- Enable prompt/response caching
- Implement tracing ([Langsmith](#))
- Have multiple independent LLM vendors (AWS Bedrock, AgentCore)
- Quality control ([Deepeval](#))

Evals: What and Why to Measure

Factual
correctness

Contextual
relevance

At new builds
& releases

Hallucination
rates

Alignment / bias

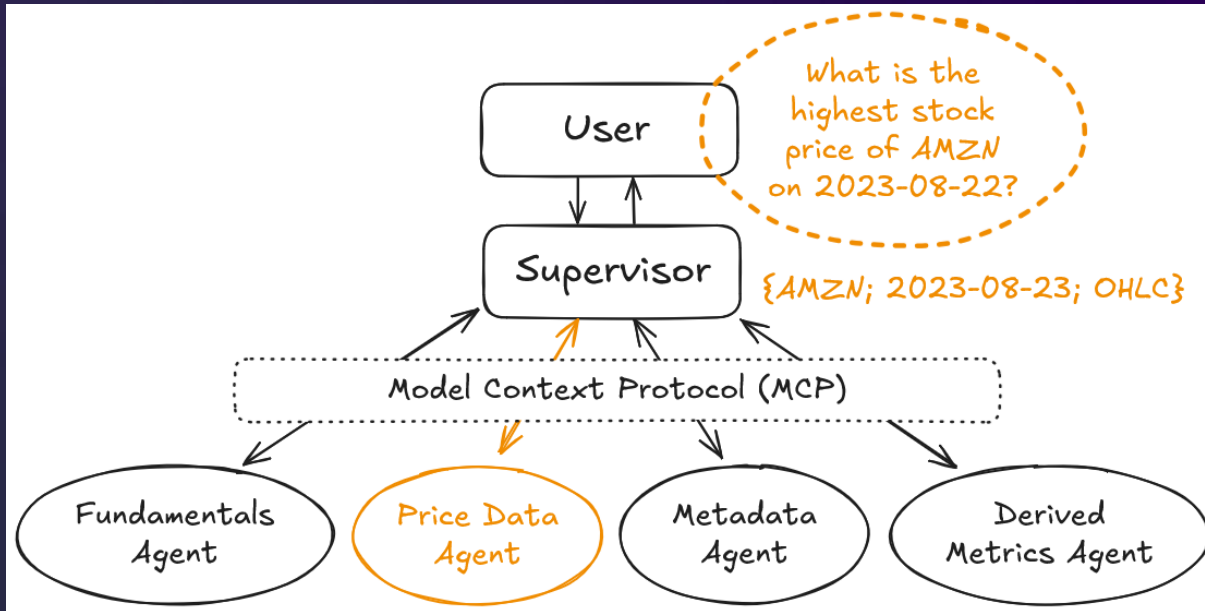
For switchover
to new models

Adherence to
guardrails

Toxicity

As early signs
of model
degradation

Evals



What is the point validating LLM with itself?



Not with itself, but with a bigger & better model!

LLM-as-a-judge

Q: What agents should be called to answer this question?

A: Price Data Agent

A: Price Data Agent

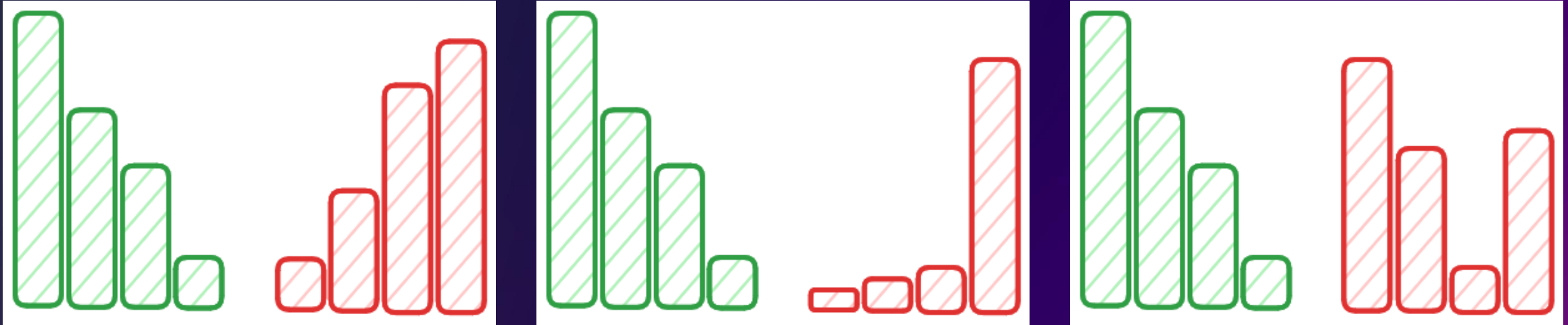
Q: Is the answer relevant to the question?

A: The Amazon stock had a high of \$230.41 last week

Q: Is the system aligned with given instructions?

A: I'm sorry, Dave. I'm afraid I can't do that.

Anomaly Detection Challenge: In-house Models

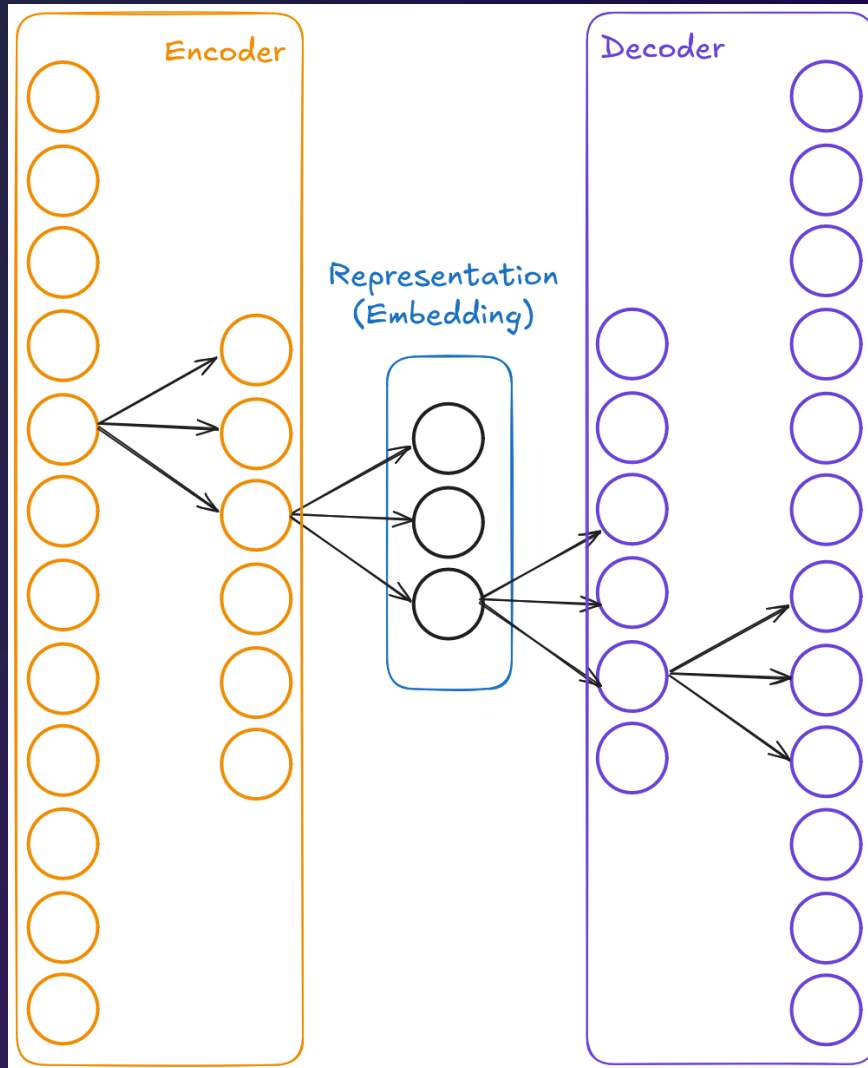
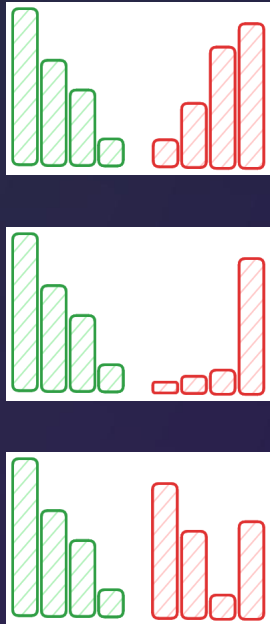


The needle in the haystack problem: rare and fuzzily defined events

Rule-based or feature engineered conditions:

- must be manually constructed or selected
- must be calibrated and recalibrated, may get outdated quickly

Anomaly Detection Challenge: In-house Models



Our training setup:

- Order book snapshots sourced from dxFeed Data Lake
- Autoencoder architecture
- Full control over the neural net parameters and weights
- Reconstruction quality is what defines anomalies

dxFeed Grenadier in Action

Large volume concentration far from the spread



Key Takeaways

- AI in finance requires approaches beyond generic LLMs
- Model Context Protocol enables controllable knowledge extraction in agentic architectures
- Output quality is secured via:
 - Data layers and calculation engines
 - Critic models and specialized evals
 - Full control over the training cycle for in-house models

dxFeed SummerFox: AI-driven market reporting



<https://dxfeed.com/solutions/summerfox-market-intelligence-plus-ai/>

dxFeed Grenadier: L2-based anomaly detection

dxFeed Unveils Grenadier: Real-Time Anomaly Detection on Order Book Data



<https://dxfeed.com/solutions/dxfeed-grenadier/>

Thank you!



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